

David Nguyen

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Portfolio available: thentd2003.github.io

Experience

Engineering Co-op

Sept. 2023 – Dec. 2024

Analytic Systems

Delta, BC

- Independently designed and simulated (LTspice/Altium) a **high-voltage flyback converter** (24VDC input), my first end-to-end power converter design, sizing the transformer, switching, and snubber components to meet output voltage and load-current targets.
- Built custom test jigs, developing and upgrading fixtures using **PCBs and 3D-printed** parts, to bench-validate the converter's output voltage and efficiency against spec using an **oscilloscope and DMM**.
- Performed **EMC/EMI compliance testing** (conducted 10kHz–10MHz, radiated 30–300MHz) on a batch of **50 power inverters** using a shielded RF enclosure, LISN, and biconical antenna; diagnosed and resolved the sole non-conforming unit, documenting per-unit conformance results for customer-facing QA.
- Enforced strict test-setup discipline across all 50 runs, covering grounding, cable routing, and chamber sealing. Setup precision directly determined pass/fail outcomes.
- Calculated individual and parallel-aggregated power ratings for high-voltage resistor banks and conducted **thermal profiling** under full-load conditions using a thermal camera, confirming safe operating margins and diagnosing airflow-driven temperature variance.

Engineering Consultant, Power Electronics

Dec. 2025 – Jul. 2026

Tech Maple Works

Burnaby, BC

- Architected a **half-bridge LCLR series-resonant induction heater**, defining the full power signal chain from DC bus through half-bridge switching, matching inductor, tank capacitor bank, and series-resonant work coil; diagnosed a gate-driver bandwidth limitation causing destructive hard-switching and resolved it via a retargeted 57–70kHz operating band.
- Bench-validated the heater against multiple physical workpieces to confirm magnetic coupling behavior against theory. Ferromagnetic steel coupled and heated efficiently at the design frequency; non-magnetic copper showed minimal heating, consistent with reduced eddy-current coupling.
- Diagnosed Curie-point-driven resonant drift as the open-loop design's core fragility and architected a **four-stage closed-loop PLL** (tank-voltage feedback, phase detector/VCO, gate-driver stage) to track resonance in real time.
- Currently designing custom **PCBs** for the full device, covering the power stage, PLL feedback, and gate driver circuitry, replacing the breadboard prototype with a production-ready layout.

Projects

Embedded Systems & Mechatronics: ESP32 Pan-Tilt Soccer/Ball Tracking Camera

[view full build →](#)

- Own the embedded/firmware side of a pan-tilt ball-tracking camera: implemented a **BLE peripheral (GATT server)** on an **ESP32** to receive real-time pan/tilt angle commands from a companion iPhone app.
- Chose a packed binary data format (**int16** angles over BLE over JSON/floats for low-latency MCU parsing, while developing the **C++** algorithm converting the ball's pixel-distance from the screen's corners into pan/tilt degrees).
- Built a **software-based encoder** in **C++** to track stepper position in firmware (no physical encoder hardware) and implemented movement-smoothing logic so the platform tracks the ball smoothly rather than jumping between raw position updates.
- Designed a bearing-mounted rotating pan base using a **bevel-gear** drive train, modeled the housing in **SolidWorks**, and **3D-printed** it along with mounting hardware; collaborating with a teammate on the iOS on-device AI ball-detection and BLE central logic.

Education

Simon Fraser University

2021 – Present

Bachelor of Applied Science in Engineering Science (Electronics)

Burnaby, BC

Simon Fraser University

May 2023 – Sep 2023

Siemens Mechatronic Systems Certification Program (Level 1)

Surrey, BC

Skills & Interests

Hardware & Lab: Soldering, Oscilloscope & DMM, PCB Design & Fabrication, **3D Printing**, Circuit Troubleshooting

Design & Simulation: Altium Designer, LTspice, KiCad, OrCAD, Proteus, MATLAB/Simulink, SolidWorks

Programming & HDL: C++, Python, VHDL, Verilog, SystemVerilog, Xilinx Vivado, Intel Quartus, ModelSim

Languages: English (Proficient), Vietnamese (Native)